Negative Goldberg: Joining the Torsion-Zero and Green-Potential Arguments

Research continuation and cross-conversation synthesis

15 October 2026

Abstract

We join five retrieved research notes from the conversation entitled "Yau 26 23 19" with the two torsion-zero continuations of 15 October. The joined identities characterize the intersection of the torsion-zero set with the double-maximal-eigenvalue locus: it consists exactly of the zeros at which the second covariant derivative of the fundamental form also vanishes. Every closed smooth torsion-zero curve with dense rank-two locus must meet this intersection. A Green identity retains the global curvature term at these points; its vanishing does not follow from their finite-order local vanishing. The unrestricted negative Goldberg conjecture remains open in these arguments.

1 The common problem and conventions

Let (M^4, g, J, Ω) be smooth, closed, connected and almost-Kähler, oriented by J, with

$$\mathrm{Ric} = -ag, \qquad a > 0, \qquad s = -4a, \qquad |\Omega|^2 = 2.$$

The target is $\nabla\Omega = 0$, without additional curvature-sign or zero-set assumptions, or an actual closed counterexample. Let $\Delta = -\operatorname{div}\nabla$, $n = \Omega/\sqrt{2}$, and

$$q = |\nabla\Omega|^2, \quad Z = \{q = 0\}, \quad \mathcal{R}_+ = W^+ - \frac{a}{3}\,\mathrm{Id}, \quad A = -\mathcal{R}_+, \quad U = \sigma_2(A) = \frac{a^2}{3} - \frac{1}{2}|W^+|^2.$$

Operator norms are Hilbert–Schmidt norms. Relative to $\mathbb{R}n \oplus n^\perp$, write

$$W^+ = \begin{pmatrix} t & v^* \\ v & -t\,\mathrm{Id}\,/2 + B \end{pmatrix}, \qquad t = -\frac{2a}{3} + \frac{q}{4}, \qquad v = \eta/\sqrt{2}, \qquad \operatorname{tr} B = 0.$$

Here $\eta = W^+(\Omega) - t\Omega$ and B is the trace-free two-dimensional block, customarily denoted W_3^+. The scalar a is not the torsion one-form used in earlier notes. We denote that one-form by ϑ:

$$\nabla\Omega = \vartheta \otimes \phi - J\vartheta \otimes J\phi, \qquad q = 4|\vartheta|^2,$$

where $|\phi|^2 = |J\phi|^2 = 2$.

2 Source correspondence and retained results

The following five complete source texts were retrieved and read. They are records of the other conversation, not independent publications.

Source and date	Mathematical content retained

`Q4_Torsion_Zero_Structure.tex`, 9 October	Second-derivative identity; complete first-derivative normal form; positive transverse indices; the scalar current retained before integration.
`Q4_Canonical_Orthogonality.tex`, 12 October	Orthogonality of η to closed real J-anti-invariant forms; positive integrating-factor criterion and its noncompact local obstruction.
`Q4_Harmonic_Product_Reduction.tex`, 13 October	Fixed-metric harmonic multiplication kernel, its exact rank-one constraint, and fourth-order norm-variance behavior near a Kähler direction.
`Q4_Determinant_Sign_Rigidity.tex`, 13 October	Rigidity under $\det \mathcal{R}_+ \le 0$ everywhere, from the Einstein Weyl equation and existence of a torsion zero.
`Q4_Green_Potential_Rigidity.tex`, 14 October	Rigidity if the Green potential of $\det \mathcal{R}_+$ is nonpositive at one torsion zero; exact weighted determinant balance.

The current conversation contributes the two 15 October notes `Goldberg_Torsion_Zeros_and_Weyl_Equation_20261015.tex` and `Goldberg_Global_Connections_20261015.tex`. Their relevant statements are the exclusion of compact clean zero curves, the normal derivative equation on a smooth zero curve, and the full coupled Weyl equation. The zero-index calculation overlaps the 9 October source and is counted once.

The source titled `Q4_Complete_Example_and_Fino_Question.tex` was located by title but not re-read for this synthesis. No additional claim about that construction or historical priority is made here. This record covers the retrieved mathematical notes, not every message of the other conversation.

For orientation, the canonical orthogonality result reads

$$\int_M \langle \eta, \alpha \rangle \, dV = 0 \quad (d\alpha = 0,\ \alpha \in \Gamma(\Lambda^-_J)).$$

The source uses the canonical-divisor theorem and the canonical curvature formula. Thus a globally smooth $f > 0$ with $d(f\eta) = 0$ implies $\eta = 0$. The compact third-Gray theorem of Apostolov–Armstrong–Drăghici [1] then implies Kählerness. Einstein Ricci is J-invariant, and $\eta = 0$ says precisely that Ω is a Weyl eigenform; these are the two conditions in that theorem. Existence of the multiplier is not supplied by this implication.

3 The common global identity, including the local zero jet

The Einstein Weyl equation is

$$\nabla^*\nabla W^+ + \frac{s}{2} W^+ - 6(W^+)^2 + 2|W^+|^2 \,\mathrm{Id} = 0.$$

Pairing it with W^+ and shifting by $a\,\mathrm{Id}\,/3$ gives

$$(\Delta + 2a)U = |\nabla A|^2 - 18 \det \mathcal{R}_+. \tag{1}$$

Indeed,

$$\Delta|W^+|^2 = -s|W^+|^2 + 36 \det W^+ - 2|\nabla W^+|^2,$$

$$|W^+|^2 = \frac{2a^2}{3} - 2U, \qquad \det W^+ = -\frac{2a^3}{27} + \frac{a}{3}U - \det A.$$

Define

$$\Psi = (\Delta + 2a)^{-1} \det \mathcal{R}_+, \qquad \mathcal{D} = (\Delta + 2a)^{-1} |\nabla A|^2.$$

The inverse exists on all scalar functions, including constants, and has a strictly positive Green kernel G_a on a connected closed manifold. Thus

$$U = \mathcal{D} - 18\Psi, \qquad \mathcal{D} \geq 0. \tag{2}$$

At $p \in Z$, the harmonic-form equation gives $\langle An, n\rangle = a$, so

$$A(p) = \begin{pmatrix} a & z^* \\ z & C \end{pmatrix}, \qquad \operatorname{tr} C = 0, \quad z = -v, \quad C = -B.$$

Consequently

$$U(p) = -|z|^2 - \tfrac{1}{2}|C|^2 = -\tfrac{1}{2}(|\eta|^2 + |B|^2)(p). \tag{3}$$

The local torsion-zero identity, established in the 9 October note and retained in the present conversation, is

$$|\nabla^2 \Omega|^2 = 4|B|^2 - 6|\eta|^2 \qquad \text{on } Z. \tag{4}$$

It follows by twice differentiating $t = \langle W^+ n, n\rangle = s/6 + q/4$ at $\nabla n = 0$:

$$\Delta t = 3|\eta|^2 - 2|B|^2, \qquad \Delta q = -2|\nabla^2 \Omega|^2.$$

Proposition 3.1 (Joined zero-point identity)**.** *Every $p \in Z$ satisfies*

$$\boxed{18\Psi(p) = \int_M G_a(p, y) |\nabla A(y)|^2 \, dV_y + \frac{1}{8}|\nabla^2 \Omega(p)|^2 + \frac{5}{4}|\eta(p)|^2.} \tag{5}$$

Proof. Combine (2)–(4); specifically $|B|^2 = |\nabla^2 \Omega|^2/4 + 3|\eta|^2/2$. □

This is a direct connection between the two existing arguments, not a second independent global rigidity theorem. The Green theorem is recovered: if $\Psi(p) \leq 0$ at one torsion zero, all terms on the right vanish. Positivity of G_a forces $\nabla A = 0$ globally. Then $U = \det A = 0$, $\operatorname{tr} A = a$, and A is a parallel rank-one operator with eigenvalues $(a, 0, 0)$. Set $\Pi = A/a$. The harmonic-form equation gives

$$\nabla^* \nabla (\Pi n) = 2\Pi(a \operatorname{Id} - A)n = 0.$$

Hence Πn is parallel, and at p it has length one. Since $|n| = 1$, this gives $n = \Pi n$ globally, hence $\nabla \Omega = 0$.

4 What the spectral intersection means exactly

Order the Weyl eigenvalues by $\lambda_1 \geq \lambda_2 \geq \lambda_3$ and set

$$\Sigma = \{\lambda_1(W^+) = \lambda_2(W^+)\}.$$

This definition uses the *largest* repeated eigenvalue.

Proposition 4.1. *On the torsion-zero set,*

$$\boxed{Z \cap \Sigma = \{p : \nabla \Omega(p) = 0, \ \nabla^2 \Omega(p) = 0\}.}$$

At every such point, $\eta = B = 0$ and $\mathcal{R}_+ = -a\, n \otimes n$.

Proof. Suppose $p \in Z \cap \Sigma$. Let λ be the repeated maximal eigenvalue of $\mathcal{R}_+$; its remaining eigenvalue is $-a - 2\lambda$. For a unit vector e in the remaining eigenline,

$$\mathcal{R}_+ = \lambda \,\mathrm{Id} - (a + 3\lambda) e \otimes e.$$

The Rayleigh value at a torsion zero is $\langle \mathcal{R}_+ n, n \rangle = -a$. The operator cannot be scalar there. Thus

$$|\langle e, n \rangle|^2 = \frac{a + \lambda}{a + 3\lambda}, \qquad \lambda \geq 0.$$

The last inequality also follows from the smallest-eigenvalue bound $-a \geq -a - 2\lambda$. Computing the off-diagonal block and the traceless restriction to $n^\perp$ gives

$$|\eta|^2 = 4\lambda(a + \lambda), \qquad |B|^2 = 2\lambda^2.$$

Equation (4) becomes

$$|\nabla^2 \Omega|^2 = -24a\lambda - 16\lambda^2.$$

Nonnegativity forces $\lambda = 0$ and $\nabla^2 \Omega = 0$, and the Rayleigh equality then gives $e = \pm n$.

Conversely, at $p \in Z$ with $\nabla^2 \Omega = 0$, the harmonic-form equation $\nabla^* \nabla \Omega = (q/2)\Omega + 2\eta$ gives $\eta = 0$. Equation (4) gives $B = 0$. The eigenvalues of W^+ are $(-2a/3, a/3, a/3)$, proving $p \in \Sigma$. □

At the intersection, (5) reads

$$\boxed{18\Psi(p) = \int_M G_a(p, y) |\nabla A(y)|^2 \, dV_y.} \tag{6}$$

It does not imply $\Psi(p) \leq 0$. If ∇A is nonzero somewhere, the right side is strictly positive. Thus "the zero set meets the spectral locus" and "the Green rigidity hypothesis holds" are separate statements. Vanishing of two derivatives is also not infinite-order vanishing for unique continuation.

5 A geometric connection supplied by closed zero curves

Let $K = \Lambda^{2,0} T^* M$ and regard torsion as a section $\sigma \in \Gamma(T^{1,0} M \otimes \bar{K})$, with the normalizations of the zero-structure notes. The canonical Hermitian connection satisfies

$$D_X \sigma = X \otimes c, \qquad X \in T^{1,0} M.$$

Let C be a closed smooth J-holomorphic curve contained in Z, with dense rank-two locus for $D\sigma$. The current conversation establishes that C is totally geodesic and that its normal torsion derivative is a smooth section

$$\zeta \in \Gamma\big(\mathrm{Hom}_{\mathbb{C}}(\bar{\nu}_C, T^{1,0} C \otimes \bar{K})\big) \simeq \Gamma(\bar{K}^{\otimes 2}|_C).$$

Here ζ is unrelated to the off-diagonal Weyl notation in some earlier manuscripts. On C, all tangent derivatives and all $(1,0)$ derivatives of σ vanish; ζ is the remaining normal $(0,1)$ derivative.

Corollary 5.1. *Every such C meets $Z \cap \Sigma$.*

Proof. The canonical curvature formula restricted to a torsion-zero curve gives

$$c_1(TM, J)[C] = \frac{s}{8\pi} \,\mathrm{Area}(C).$$

As $c_1(\bar{K}) = c_1(TM, J)$,

$$\deg(\bar{K}^{\otimes 2}|_C) = \frac{s}{4\pi}\,\mathrm{Area}(C) < 0.$$

A nowhere-zero smooth section would trivialize this complex line bundle and force degree zero. Hence $\zeta(p) = 0$ somewhere. All of $D\sigma$ then vanishes at p. Since $\sigma(p) = 0$, the canonical and Levi-Civita derivatives agree there, and the torsion identification gives $\nabla^2\Omega(p) = 0$. Proposition 4.1 applies. □

This corollary requires neither isolated zeros of ζ nor a positivity-of-indices assertion. It extends the earlier clean-curve exclusion by identifying where a rank drop must occur spectrally. It does not assert that every component of the torsion-zero set is a smooth curve.

The equation retained on C is $D^{1,0}\zeta = r$, with r a normal derivative of c, and $D^{1,0}r = 0$ under the stated dense rank-two hypothesis. It does not say that ζ is a closed J-anti-invariant two-form on M. The canonical-divisor theorem used in the orthogonality note therefore cannot be applied to this section by changing its name.

6 The resulting common continuation point

The merged statements are compatible and give the following precise restrictions on any non-Kähler closed candidate:

1. $Z \neq \varnothing$, by the classical Nijenhuis-bundle obstruction $c_2(T^{1,0}M \otimes \bar{K})[M] = 5\chi + 6\tau > 0$.
2. $\Psi(p) > 0$ at every $p \in Z$, with its exact value prescribed by (5). In particular $\det \mathcal{R}_+ > 0$ somewhere.
3. At a torsion zero, membership in Σ is exactly second-order torsion vanishing. Any closed smooth zero curve of the preceding type must contain such a point.
4. Neither that intersection nor the finite-order vanishing removes the global integral in (6).

Accordingly, the direct unresolved connection on this route is a restriction on the actual Einstein–almost-Kähler system that forces the nonnegative integral in (6) to vanish, or that forces the Green sign at a torsion zero without a zero-curve hypothesis. The present identities do not furnish that restriction. A valid proof may use a different route; none of these auxiliary conditions is added to the original conjecture.

The harmonic multiplication formulation remains an exact alternative for a fixed metric, not an additional curvature inequality. Likewise, the existing full B equation retains first-order coupling to η; it cannot be replaced by a homogeneous positive-potential equation. These observations preserve both conversations' usable results while keeping the remaining implication mathematically explicit.

References

[1] V. Apostolov, J. Armstrong and T. Drăghici, *Local models and integrability of certain almost Kähler 4-manifolds*, Math. Ann. 323 (2002), 633–666. `https://arxiv.org/abs/math/0109098`. Lemma 1 and Theorem 2.

[2] A. Derdziński, *Self-dual Kähler manifolds and Einstein manifolds of dimension four*, Compositio Math. 49 (1983), 405–433. `https://archive.numdam.org/item/CM_1983__49_3_405_0/`.

[3] C. LeBrun, *Einstein metrics, symplectic minimality, and pseudo-holomorphic curves*, Ann. Global Anal. Geom. 28 (2005), 157–177. `https://arxiv.org/abs/math/0409290`. Proposition 2.1.

[4] The seven research notes identified in Section 2. The Green argument is inherited from the other conversation; the closed-zero-curve argument is inherited from the present conversation. Propositions 3.1 and 4.1 and Corollary 5.1 state their explicitly derived connections.